28-02-2026

Agenda:

pandas

Git

Group-1	Group-2	Group-3	Group-4
N1	N4	N5	N6
N2	↓	↓	↓
N3	attributes	attribute	attributes
↓			
attributes			

→ Dataset

6 → attributes → groups

person-7 → attributes

ML-model (learn based on 6 student) → 1, 2, 3, 4

classification

(1, 1, 1) → group-1 → 1

group-2 → ML → 2

group-3 → 3

group-4 → 4

encoding

group → categorical (numerical)

"A"

"B" → character / string → pandas → D → A

"C"

"D"

A 2

D
C
B
A

16

"First" "Second" "Third"

1 2 3

2 > 1 : True

"Two" > "First" : True

String → converted → dtype → to handle order
↓
logic

event	date	ticket_id	price
Concert	2026-02-02	—	—
—	—	—	—
—	—	—	—
conference	—	—	—

Data analysis → concert →

rows → date → 2026-02-02

string → year
→ month
→ day
→ weekday

row1 → 02-02-2026 ⟷ 28-02-2026

str

Jan-1 → new year

→ today Jan-30

new year — today → —

holi — today → +

python + OOP (DS)

Numpy (DS)

pandas (DS)

Git (developer)

Data visualization

tools of data science

DS :— understanding data →

→ data —

→ analysis →

→ fix →

→ getting stats →

→ plotting →

Git :— Tools to manage code

uv → project + dependencies

↓

env

↓

libraries

↓

python

git → project → p1 ← code
→ p2 ← code
→ p3 ← code
→ p4 ← code

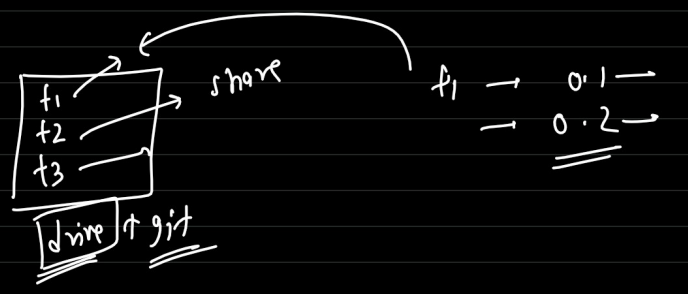
p1 → Today → + 10000 lines → working → save → commit cp
→ Day +1 → + 10 lines → error
up to 10000 lines → error → save → commit cp
→ correct
git

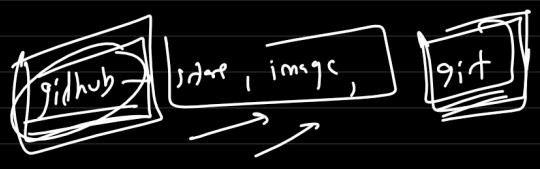
git → diff → file

Version : 0.1
→ : 0.2
→ 0.3

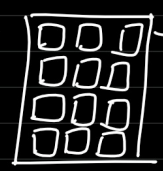
code → git ^ version control tool

git → tools
github → central repositor of your code + git





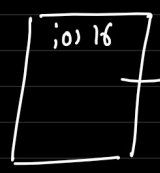
git + online

project → ios 16 →  → consumer

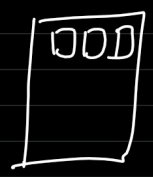
millions of people
ios 16 → project → copy
↓
work

main → branch

enhance → camera_manual → branch



copy



→ test → error → ✓

ios 16 → update

(branch) def add (a,b):
print (a+b)

→ milliy

↓

main branch

enhancement: (branch)

D1 – camera

copy

def add (*args):
for i in ar

↓

testing

D2 → widget

D3 →

update the code

merge

Git:

→ manage code by creating checkpoints, version

→ seperate code by branches

main → (panda) → 0.0.2

↓

git

→ branch → 0.0.3

numpy → 0.0.16

↓

git

branch

online : repo → github
github

D1:

local → git
code

↑ git init
→ git add .
→ git commit -m "abc"
→ git branch -M main
→ git remote add origin
url of the repo
→ git push -u origin main

1st time thing

D2:

git clone url-of-repo

D2: → uploaded files using github UI → .txt → github → online

github online	local repo
f1	f1
f2	→ git pull origin branch-name
	↓
	main

D1 → D2 → main → a.py, b.py → 9 pm → 6 pm

log off
↓
log in
↓
git pull

project-1

study-1

pandas

numpy

.venv

.py

pandas

.venv

.py